KK-B-151f
March 2, 1967

SUPERSEDING
Int. Fed. Spec. KK-B-00151e(YD)
August 13, 1965 and
Fed. Spec. KK-B-151d
July 8, 1960

BELT, SAFETY, INDUSTRIAL, AND STRAP, SAFETY, INDUSTRIAL: LINEMAN'S TYPE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for a nylon neoprene safety belt and safety strap.

1.2 Classification.

1.2.1 *Type.* The belt and strap shall be of the following type, as specified (see 6.2).

Type:
I—Industrial safety belt.
II—Industrial safety strap.

1.2.2 *Sizes.*

1.2.2.1 Type I belts shall be furnished in the following sizes, as specified (see 6.2):

Size:
34—approximate 34-inch waist, D size 19.
36—approximate 36-inch waist, D size 20.
38—approximate 38-inch waist, D size 21.
40—approximate 40-inch waist, D size 22.
42—approximate 42-inch waist, D size 23.
44—approximate 44-inch waist, D size 24.
46—approximate 46-inch waist, D size 26.
48—approximate 48-inch waist, D size 28.

1.2.2.2 Type II strap shall be 70 inches in length.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

V-T-295—Thread, Nylon.
FF-R-556—Rivets, Solid, Small; Rivets, Split, Small; Rivets, Tubular, Small; and Caps, Rivets; General Purpose.
QQ-P-416—Plating, Cadmium (Electrodeposited).
QQ-S-781—Steel, Strapping, Flat.
QQ-Z-325—Zinc Coating, Electrodeposited, Requirements For.
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-576—Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585—Boxes, Wood Wirebound.
PPP-B-591—Boxes, Fiberboard, Wood-Cleated.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.
PPP-B-665—Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676—Box, Setup.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 751—Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and

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Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.

MIL-B-10377—Boxes, Wood-Cleated, Veneer, Paper-Overlaid.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials (ASTM) Standards:

A 164-55—Electrodeposited Coating of Zinc on Steel.

A 165-55—Electrodeposited Coatings of Cadmium on Steel.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, Pa., 19103).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), the contractor shall furnish, within the time limit specified in the contract or order, a preproduction sample for inspection. This inspection, which shall consist of the examination and tests specified in 4.5 and 4.6.6, shall be conducted prior to the submission for acceptance of the remainder of the safety belts and straps furnished under the contract. The preproduction sample shall be identical to the units the contractor proposes to furnish in fulfillment of the contract. Approval of the preproduction sample on the basis of the specified inspection shall not relieve the contractor of his obligation to meet the requirements of the specification for the remaining units delivered under the contract.

3.2 Material. Material shall be as specified herein. Material not definitely specified shall be of the quality used for the purpose in commercial practice and shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.2.1 *Nylon strap.* The nylon strap shall be nylon fabric of plied construction, impregnated and coated with neoprene, and of a color furnished commercially by the manufacturer with the center ply or plies of a contrasting color. The strap shall have a minimum width of 1–3/4 inches, a minimum thickness of four ply, and a minimum breaking strength, without buckle hole, of 5500 pounds.

3.3 Dee rings, snaphooks, and buckles. Dee rings, snaphooks, and buckles shall be steel drop forgings or welded parts and shall be zinc coated or cadmium plated as specified (see 6.2). Zinc coating shall be in conformance with QQ-Z-325, type II, class 1 or ASTM A 164, type GS. Cadmium plating shall be in conformance with QQ-P-416, type II, class 1 or ASTM A 165, type NS.

3.3.1 *Dee rings.* Dee rings shall be as indicated in figure 1 and shall be so proportioned as to eliminate the possibility of rollout and accidental disengagement of a snaphook caused by depression of the snaphook keeper latch by the dee ring. Each dee ring shall have installed a suitable copper or steel safety liner to prevent wear between the bar of the dee ring and the fabric enclosing it. Dee rings shall be capable of withstanding a tensile load of 1500 pounds

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without deforming more than 1/64 inch when tested as described in 4.6.3.1. No cracks or fractures shall be visible in the dee rings when subjected to the test described in 4.6.3.2.

3.3.2 *Snaphooks.* Snaphooks shall be of the round nose type with closed stirrup essentially as detailed in figure 1. The round nose shall overlap the keeper latch in such manner as to prevent the latch from being accidentally engaged by the dee ring to release the snap. Snaphooks shall be capable of withstanding a tensile load of 1500 pounds without deforming more than 1/32 inch when tested as described in 4.6.4.1. No cracks or fractures shall be visible in the snaphooks when subjected to the test described in 4.6.4.2.

3.3.2.1 *Keeper.* Keeper shall fit snugly, without binding, between the sides of the nose of the hook and shall extend into the round nose housing in the direction of the long axis of the keeper not less than 5/32 inch.

3.3.3 *Buckles.* Buckles shall be as indicated on figure 1 and shall be single or double-tongued.

3.3.3.1 *Buckle frames.* Buckle frames shall be capable of withstanding a tensile load of 1500 pounds without deforming more than 1/64 inch when tested as described in 4.6.5.1. No cracks or fractures shall be visible in the buckle frames when subjected to the test described in 4.6.5.2.

3.3.3.2 *Buckle tongues.* No cracks or fractures shall be visible in the buckle tongues when subjected to the test described in 4.6.5.2.

3.4 Rivets and burrs. Unless otherwise specified (see 6.2), rivets and burrs shall be of the solid copper, belt type not less than 1/8 inch in diameter conforming to FF-R-556.

3.5 Thread. Nylon thread conforming to V-T-295, type XI, shall be used for stitching. The thread shall have a breaking strength of not less than 40 pounds.

3.5.1 *Stitching.* Stitch type 301, Fed. Std. No. 751, shall be used. Not less than 5 nor more than 7 stitches per inch shall be allowed.

3.6 Type I, industrial safety belt. The industrial safety belt shall be of nylon-neoprene design in accordance with figure 2. Buckles shall be single tongued. Component parts shall not fracture or rupture when subjected to the test described in 4.6.6.1. The belts shall conform to the dimensional requirements shown in table I.

3.6.1 *Miscellaneous belt parts.* Unless otherwise specified (see 6.2), the following belt parts shall be furnished.

(a) Knife snap.
(b) Tape thong.
(c) Ring for glove pouch.
(d) Pocket tabs.
(e) Plier pouch.

3.7 Type II, industrial safety strap. Nylon safety straps shall be in accordance with figure 3. Nylon safety straps shall be of one size only, adjustable by means of a

Table I. *Dimensions for industrial safety belt*

Dimensions in inches (see fig. 2)						
Size	A±1/2	B	C±1/4	D±1/4	E	F
34	43–1/2	34	19	7–1/2	17	2–1/2
36	45–1/2	36	20	8	17–1/2	3–1/2
38	47–1/2	38	21	8–1/2	18	4
40	49–1/2	40	22	9	18–1/2	4–1/2
42	51–1/2	42	23	9–1/2	19	5
44	54–1/4	44	24	10	19–1/2	4–3/4
46	56–1/4	46	26	10	20–1/4	4–3/4
48	59	48	28	10	21	4

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buckle. Component parts shall not fracture or rupture when subjected to the test described in 4.6.6.1.

3.8 **Instructions for care of belt and strap.** The manufacturer shall include with each belt and strap, written instructions for care of the belt and strap during the period of their use. Instructions shall recommend special treatment that may be required to withstand the effects of climatic tests involving rapid changes in humidity as well as the cold and heat normally encountered in storage.

3.9 **Marking.**

3.9.1 *Other requirements.* Each body belt and strap shall have a suitable metal marker attached on which the manufacturer's name, the month and year of manufacture, and the manufacturer's serial number shall be permanently stamped.

3.10 **Workmanship.** Workmanship shall be as specified herein, and in accordance with the quality of workmanship customarily practiced by the contractor to attain maximum efficiency and safety as determined by commercial usage.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Preproduction sample.** When specified (see 6.2), the preproduction sample shall be tested at the place of manufacture. Examination and tests shall be in conformance with 4.5 and 4.6.6.

4.3 **Lot.** A lot shall consist of all belts and straps of the same type and size offered for delivery at the same time, plus the quantity required for destructive tests.

4.4 **Sampling.**

4.4.1 *Sampling for examination.* Sampling for examination of critical defects does not apply since all belts and straps are required to be examined for critical defects (see table II). Sampling for examinations for minor defects (see table II), shall be in accordance with MIL-STD-105, with lot acceptance or rejection based on an acceptable quality level (AQL) of 2.5 percent defective.

4.4.2 *Sampling for tests.* Sampling for all applicable tests shall be in accordance with MIL-STD-105, based on inspection level S-2 and an AQL of 4.0 percent defective. For the nylon webbing test, three samples shall be taken.

4.5 **Examination.** Each unit presented for acceptance shall be visually and dimensionally examined for critical defects listed in table II, or for any other defects that may result in hazardous or unsafe conditions for the user. Samples selected in accordance with 4.4.1 shall be examined for minor defects listed in table II.

Table II. *Classification of defects in accordance with MIL–STD–105*

Categories	Defects
Critical	
01	Less than minimum specified width or thickness for belt or strap.
02	Poor riveting, not properly peened and drawn.
03	Broken or incorrect stitching, except on holster.
04	Incorrect dimension (where dimensions are specifically toleranced).
05	Incorrect assembly.
06	Missing, loose, cracked, broken, or distorted parts.
07	Delamination of nylon plies.
Minor	
201	Poorly peened rivets, not properly peened and drawn.
202	Rust on metallic parts or poor plating.
203	Two sixty-fourths of an inch or more below minimum thickness for holster.
204	Inadequate coating of nylon.
205	Blisters in coating.

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4.6 Tests.

4.6.1 *Nylon webbing.* The three samples of the nylon fabric shall be tested for tensile breaking strength. Each sample shall be secured in the testing machine and a total load of 5500 pounds applied with a steady pull at a rate of 4 inches per minute. If either sample shows signs of breaking, the entire lot shall be rejected.

4.6.2 *Thread.* Thread shall be tested in conformance with V-T-295.

4.6.3 *Dee rings.* Samples selected in accordance with 4.4.2 shall be tested as detailed below.

4.6.3.1 *Tensile load.* A metal clasp approximately 1-1/2 inches wide shall be attached to the base of the dee ring and the clasp inserted in one set of jaws of the testing machine. The curved portion opposite the base shall be enclosed in a wire loop, the loose ends of the wire inserted in the other set of jaws of the testing machine, and a total load of 1500 pounds applied to the dee ring. After release of the load, the dee ring shall be examined for any indication of failure (see 3.3.1).

4.6.3.2 *Cracking.* A load shall be applied on the base and opposite curved portion of the dee ring, either in a testing machine or a hand vise, sufficient to close the jaws of the dee ring to a distance of approximately one-half of the original distance of the opening. After releasing the load, the dee ring shall be examined for cracks or fracture (see 3.3.1).

4.6.4 *Snaphooks.* Samples selected in accordance with 4.4.2 shall be tested as specified following.

4.6.4.1 *Tensile load.* A metal clasp shall be attached on the barrel roller around the bar in the stirrup of the snaphook and the clasp inserted on one set of jaws of the testing machine. The eye of the hook shall be enclosed in a wire loop, the loose ends of the wires inserted in the other set of jaws of the testing machine, and a total load of 1500 pounds applied to the snaphook. After release of the load, the snaphook shall be examined for any indication of failure (see 3.3.2).

4.6.4.2 *Cracking.* The bill of the hook shall be bent outward approximately 30° in the plane of the shank. This may be conveniently done with a pipe slipped over the bill of the hook and resting on the base of the eye. The straight portion of the shank shall be bent through an angle of 180° over a mandrel one-half inch in diameter. After the shank has been bent, the hook shall be examined for cracks and any indication of fracture (see 3.3.2).

4.6.5 *Buckles.* Samples selected in accordance with 4.4.2 shall be tested as specified below.

4.6.5.1 *Tensile load.* Metal clasps, attached to opposite sides of the buckle frame, shall be inserted in the jaws of the testing machine and a total load of 1500 pounds shall be applied to the buckle. After releasing the load, the buckle shall be examined for any indication of failure (see 3.3.3.1).

4.6.5.2 *Cracking.* Any portion of the buckle or any tongue shall be bent through an angle of 180° and examined for cracks or indications of fracture (see 3.3.3.1 and 3.3.3.2).

4.6.6 *Drop tests.* Preproduction samples, when specified, and samples of complete belts or straps selected in accordance with 4.4.2, shall be tested as specified below.

4.6.6.1 *Lineman's belt and safety strap.* When lineman's belts are procured with safety straps, the belt and strap shall be tested together. When the belts or straps are procured separately, the item under procurement shall be tested as described below with a strap or belt provided by the supplier. The body belt shall be securely fastened around a 250-pound bag of sand. The safety strap shall be passed over a securely anchored 2-inch diameter bar and fastened, by means of snaphooks, to the dee rings of the body belt. The bar shall be positioned at such a height that the sandbag will be suspended at least 2 feet above the ground or floor. The sandbag shall then be raised and permitted to fall free a distance of 2 feet before being stopped by the belt and strap. This shall be repeated for a second drop. After the second

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drop, the component parts of the belt or strap shall be examined for any fracture or rupture.

4.7 Inspection of preparation for delivery. The packing and marking of the belts and straps shall be inspected to determine conformance with the applicable requirements of section 5. The packing and marking requirements of section 5 identified to referenced specifications and standards shall be inspected in accordance with the applicable requirements of these documents. The packing requirements of section 5 not identified to referenced specifications shall be examined for the defects listed in 4.7.1, in accordance with MIL-STD-105 using an AQL of 4.0 percent defective.

4.7.1 *Classification of defects.*

Major	
101	Closure of shipping container not as specified.
102	Strapping not as specified.
103	Gross weight of shipping container exceeds limit specified.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A.* Each complete belt or strap shall be packaged method IC-2 in accordance with MIL-P-116. Containers shall be of a size which will snugly contain the belt or strap in minimum cubage.

5.1.2 *Level B (civil agencies).* Each complete belt or strap shall be packaged in a close-fitting box conforming to PPP-B-566; PPP-B-636, class domestic; PPP-B-665; or PPP-B-676. The fiberboard box shall be closed in accordance with method I of the appendix to the box specification.

5.1.3 *Level C.* The belts or straps shall be packaged to afford protection against corrosion, deterioration, and damage from the supplier to the initial destination. The supplier may use his commercial practice provided it fulfills the above requirements.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Packaged belts and straps of identical type and size shall be packed in close-fitting boxes conforming to PPP-B-585, class 3; PPP-B-591, class II; PPP-B-601, overseas type; PPP-B-621, class 2; PPP-B-636, class weather-resistant; or PPP-B-576, type 2 load. Boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. Flat steel strapping shall be type I, class B, or QQ-S-781. The gross weight of wood and wood-cleated boxes shall not exceed 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the box specification.

5.2.2 *Level B.*

5.2.2.1 *Military.* Packaged belts and straps of identical type and size shall be packed in close-fitting boxes conforming to PPP-B-585, class 1; PPP-B-591, domestic type; PPP-B-601, domestic type; PPP-B-621, class 1; and PPP-B-636, class domestic. Boxes shall be closed and strapped in accordance with the appendix to the applicable box specification. The gross weight of wood and wood-cleated boxes shall not exceed 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the box specification.

5.2.2.2 *Civil agencies.* Twelve unit packages of belts or straps shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with method I of the appendix to the box specification.

5.2.3 *Level C.* The belts or straps shall be packed to insure carrier acceptance and safe delivery at destination at the lowest rates, in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Standard packaging (civil agencies). The standard packaging for civil agencies shall be as specified in 5.1.2 and 5.2.2.2, respectively.

5.4 Marking (see 6.2).

5.4.1 *Civil agencies.* In addition to any special marking required by the contract or

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order, shipping container shall be marked in accordance with Fed. Std. No. 123.

5.4.2 *Military agencies.* In addition to any special marking required by the contract or order, shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The safety belt and strap covered by this specification are intended for use by linemen or electricians for the installation or repair of transmission or communication lines.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Type required (see 1.2.1).
(*c*) Size of belt required (see 1.2.2).
(*d*) Preproduction sample if required (see 3.1).
(*e*) Type of coating required (see 3.3).
(*f*) Type of rivets and burrs other than specified (see 3.4).
(*g*) When miscellaneous parts are not required (see 3.6.1).
(*h*) Testing of preproduction sample (see 4.2).
(*i*) Selection of the applicable levels of packaging and packing required (see 5.1 and 5.2).
(*j*) Special marking required (see 5.4).

6.2.1 *Standard pack for civil agencies.* The standard pack requirements in 5.3 are intended for use in procurements of stores stock replenishments. Procuring officers should use the standard pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot or intermediate storage point for temporary storage, subsequent issue or shipment to eventual user.

6.3 Classification changes. This document differs from the superseded specifications in respect to the following classification:

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Type I	Type I	Type I	Type I
----	Model A	Model A	Model A
----	----	----	Class 1
----	Style L	Class 1	Class 2
----	----	Class 2	Class 3
----	Style N	----	----
----	----	Model B	Model B
----	----	Model C	Model C
Type II	Type II	Type II	Type II
----	Style L	Material L	Material L
----	Style N	Material N	Material N
	----	Type III	Type III

MILITARY INTEREST:

CUSTODIANS:

Navy—YD

Army—MU

Air Force—84

Review Activities:

Army—EL, MU

Navy—YD, MC

Air Force—11

User Activities:

Army—MO

Navy—SH, CG

Preparing Activity:

Navy—YD

CIVIL AGENCY INTEREST:

GSA—FSS

Com—NBS

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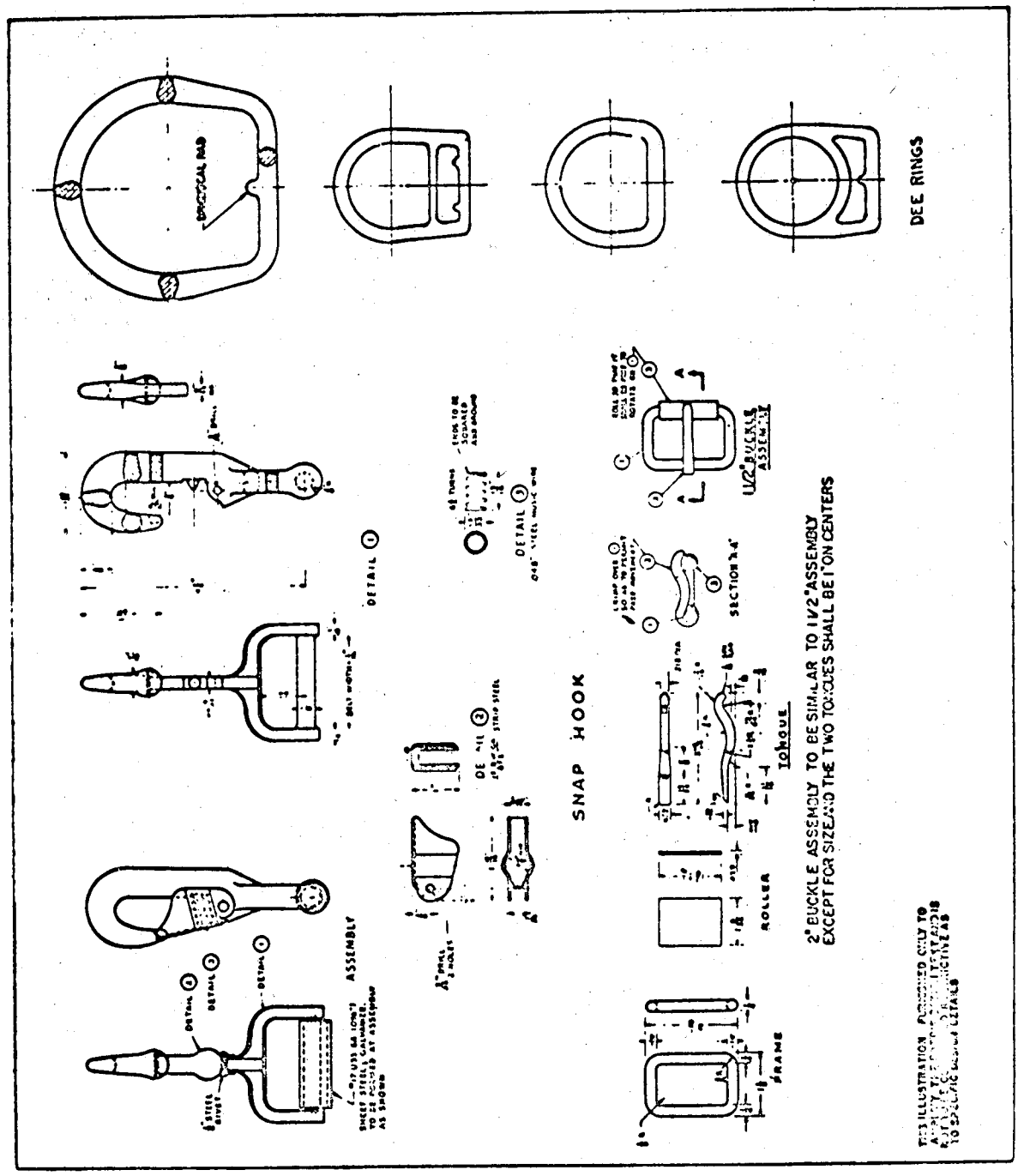


FIGURE 1

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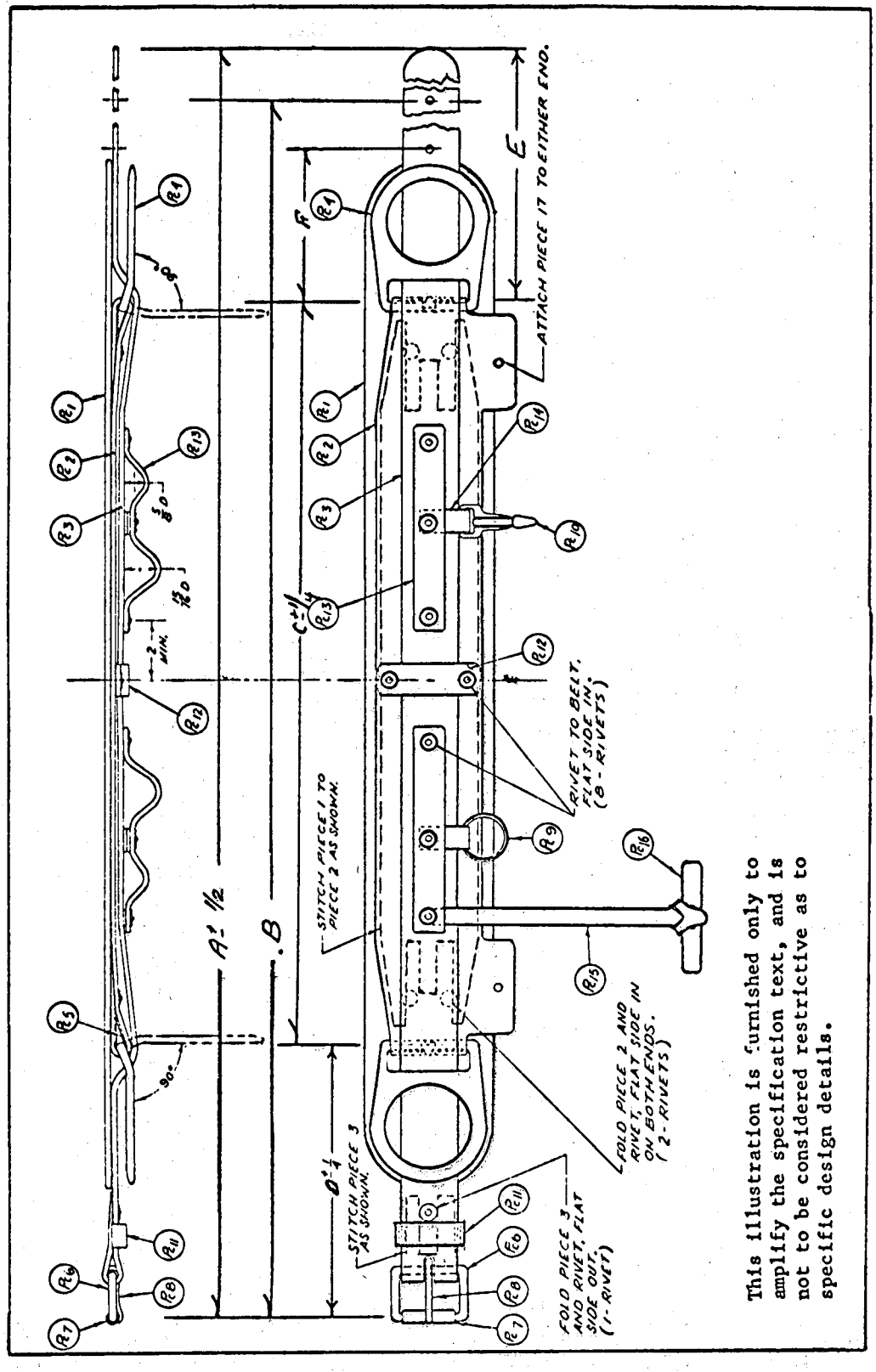


FIGURE 2. Industrial safety belt.

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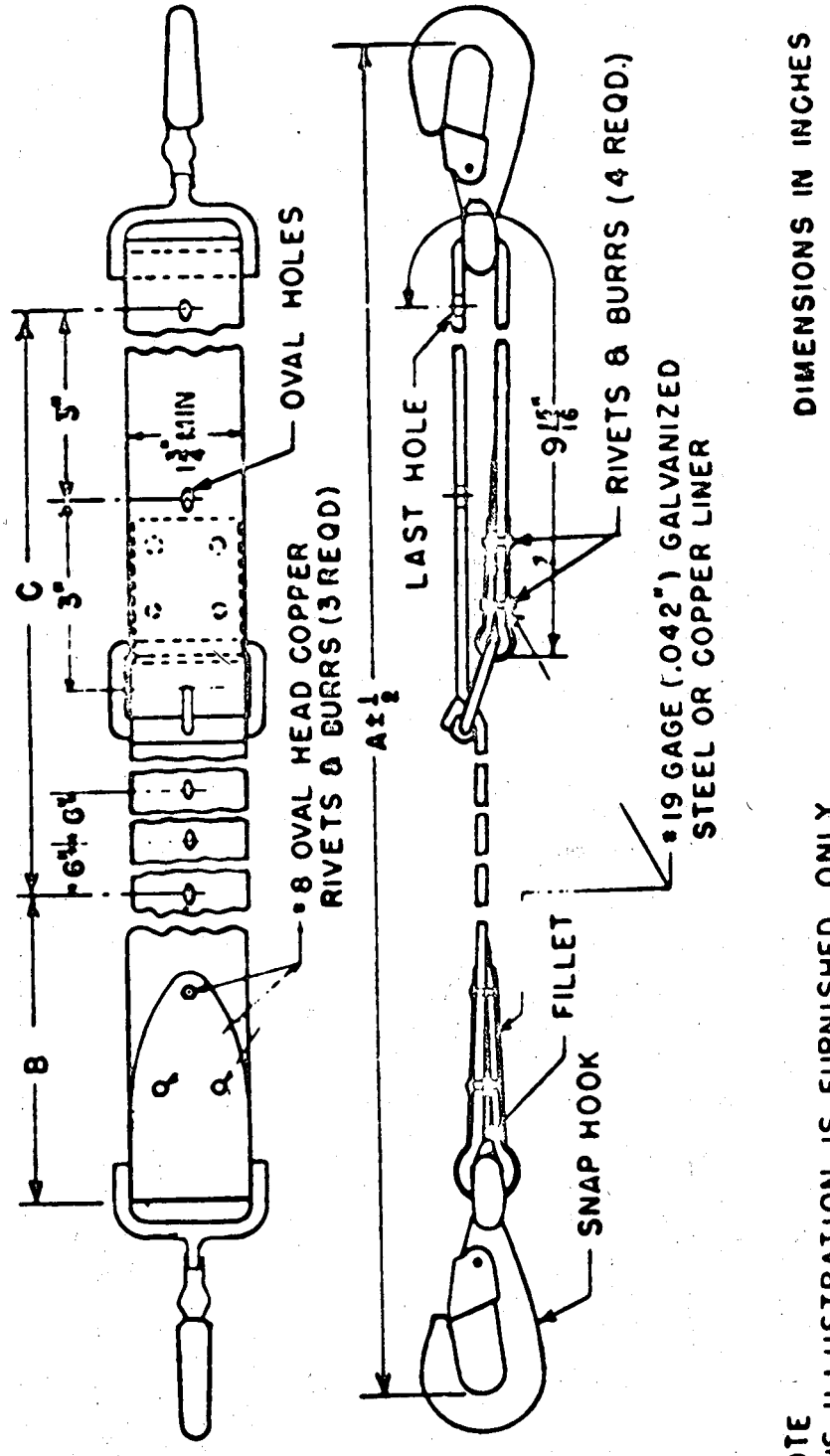


FIGURE 3. Safety strap not to scale.

Orders for this publication are to be placed with acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price cents each.

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